

Backend Error Handling and Troubleshooting

Backend diagnostics, recovery and escalation.



This document explains its subject first in plain language, then in engineering detail. It is grounded in the Craves repository snapshot and deliberately separates current implementation, legacy/reference code, milestone foundations and repository gaps so readers do not mistake aspiration for proof.

Version 1.0 | Evidence date: 14 August 2026 | Repository: [rmorampudi09-arch/Craves-Build-platform](#) | Snapshot commit: [3225f7fa531a6b07185fb5e034f7930f8bd8b571](#)

triage model - plain-language purpose

Plain-English explanation

Craves includes a governed semantic meal concierge for assisted discovery rather than an authority that can bypass commerce rules.

How it works technically

Recent main history and [apps/customer-web-next/docs/signalr-ai-concierge.md](#) describe SignalR semantic concierge work. Recommendations still rely on governed product data and ordinary catalog, quote, authorization and order paths for transactions.

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Errors, edge cases and controls

- Stale UI/cache must not override server state.
- UI visibility is not authorization evidence.
- Retries of money/state-changing operations require idempotency awareness.
- Provider/configuration outages should preserve core domain consistency where safe.
- Missing repository proof is reported as a gap rather than converted into a production claim.

Diagnostic and validation steps

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Data, security and deployment impact

The ai boundary owns its authoritative state; clients request or display changes but do not become the source of truth. Secrets and unnecessary personal data stay outside logs and PDFs. Releases must keep API, policy, schema and artifact compatible, with health/readiness evidence and rollback/backout appropriate to the change.

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A useful failure model separates user input/state conflicts, identity/permission failures, dependency outages, data faults and deployment/configuration incidents.

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triage model - deployment, security and operational validation

Plain-English explanation

The repository has broad CI/CD references across backend, frontend, mobile, APIM, data, privacy, security, smoke and release controls.

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customer-web-next README describes ACR build/push, Container App revision deployment, readiness checks and restoration of the prior image/revision on failed readiness. Pipeline names not resolvable at their cited path are classified as reference-only instead of assigned invented triggers.

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1. Capture time, environment, operation, expected/actual result and a user-safe identifier. 2. Capture HTTP status, stable error code and request/correlation ID; never copy OTPs, passwords, access tokens or provider secrets. 3. Verify caller role/ownership and current authoritative state before retrying. 4. Check owning service health, then its provider/dependency after local/configuration causes are excluded. 5. If a release is implicated, compare artifact/config marker with prior known-good revision and validate rollback compatibility.

Data, security and deployment impact

The errors boundary owns its authoritative state; clients request or display changes but do not become the source of truth. Secrets and unnecessary personal data stay outside logs and PDFs. Releases must keep API, policy, schema and artifact compatible, with health/readiness evidence and rollback/backout appropriate to the change.

Evidence status

Implemented on main. Evidence: services/order-service/README.md; services/notification-service/README.md; apps/customer-web-next/README.md. Snapshot: main @ 3225f7fa531a (14 August 2026).

403 Forbidden - deployment, security and operational validation

Plain-English explanation

The repository has broad CI/CD references across backend, frontend, mobile, APIM, data, privacy, security, smoke and release controls.

How it works technically

customer-web-next README describes ACR build/push, Container App revision deployment, readiness checks and restoration of the prior image/revision on failed readiness. Pipeline names not resolvable at their cited path are classified as reference-only instead of assigned invented triggers.

Flow described in words

1. Actor starts the capability with the appropriate identity/context. 2. Client/BFF validates local prerequisites and sends an API request. 3. APIM and the owning service enforce identity, role, ownership and business-state rules. 4. The domain service reads/writes authoritative state and calls providers only through defined boundaries. 5. The caller receives authoritative result or a traceable error; support uses request/correlation evidence rather than secrets.

Errors, edge cases and controls

- Stale UI/cache must not override server state.
- UI visibility is not authorization evidence.
- Retries of money/state-changing operations require idempotency awareness.
- Provider/configuration outages should preserve core domain consistency where safe.
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Evidence status

Implemented on main. Evidence: apps/customer-web-next/README.md; repository YAML search; docs/handover/. Snapshot: main @ 3225f7fa531a (14 August 2026).

expired token - plain-language purpose

Plain-English explanation

Craves includes a governed semantic meal concierge for assisted discovery rather than an authority that can bypass commerce rules.

How it works technically

Recent main history and apps/customer-web-next/docs/signalr-ai-concierge.md describe SignalR semantic concierge work. Recommendations still rely on governed product data and ordinary catalog, quote, authorization and order paths for transactions.

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Evidence status

Implemented on main. Evidence: apps/customer-web-next/docs/signalr-ai-concierge.md; commit 71e795b. Snapshot: main @ 3225f7fa531a (14 August 2026).

expired token - technical architecture

Plain-English explanation

Authentication proves who the caller is; authorization decides what that caller may do. Craves uses phone OTP for the current web and JWT-based authorization in backend services, with Entra External ID/PKCE documented for the mobile milestone.

How it works technically

Secure HTTP-only cookies keep web token state away from ordinary browser JavaScript. Backend services evaluate roles such as USER, CHEF and ADMIN and may also enforce object ownership. Mobile milestone guidance stores refresh credentials in Keychain/Keystore and access tokens in memory.

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Evidence status

Implemented on main. Evidence: apps/customer-web-next/README.md; services/user-chef-service/README.md; services/order-service/README.md; docs/handover/2026-07-30-customer-mobile-auth-foundation.md. Snapshot: main @ 3225f7fa531a (14 August 2026).

expired token - end-to-end flow

Plain-English explanation

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missing identity context - plain-language purpose

Plain-English explanation

Craves includes a governed semantic meal concierge for assisted discovery rather than an authority that can bypass commerce rules.

How it works technically

Recent main history and [apps/customer-web-next/docs/signalr-ai-concierge.md](#) describe SignalR semantic concierge work. Recommendations still rely on governed product data and ordinary catalog, quote, authorization and order paths for transactions.

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Implemented on main. Evidence: [apps/customer-web-next/docs/signalr-ai-concierge.md](#); commit 71e795b. Snapshot: main @ 3225f7fa531a (14 August 2026).

missing identity context - technical architecture

Plain-English explanation

A useful failure model separates user input/state conflicts, identity/permission failures, dependency outages, data faults and deployment/configuration incidents.

How it works technically

order-service documents Problem Details fields and codes including ORDER_NOT_FOUND, ORDER_IDEMPOTENCY_CONFLICT, INVALID_STATUS, CANCELLATION_NOT_ALLOWED, REFUND_FAILED, UNAUTHORIZED, FORBIDDEN, BACKEND_CHANGE_DISABLED and BACKEND_UNAVAILABLE. Diagnostics should start from request/correlation evidence.

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chef state mismatch - plain-language purpose

Plain-English explanation

Chef mode is the supply side of the marketplace: onboarding, verification, menus, kitchen/order operations and earnings.

How it works technically

user-chef-service persists chef profiles and verification state. The frontend can switch context, but the server remains the authority: showing Chef mode never grants a CHEF role by itself.

Flow described in words

1. Actor starts the capability with the appropriate identity/context. 2. Client/BFF validates local prerequisites and sends an API request. 3. APIM and the owning service enforce identity, role, ownership and business-state rules. 4. The domain service reads/writes authoritative state and calls providers only through defined boundaries. 5. The caller receives authoritative result or a traceable error; support uses request/correlation evidence rather than secrets.

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Evidence status

Implemented on main. Evidence: services/user-chef-service/README.md; apps/customer-web-next/src/components/chef/; apps/customer-web-next/src/components/chef-mode/. Snapshot: main @ 3225f7fa531a (14 August 2026).

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1. Capture time, environment, operation, expected/actual result and a user-safe identifier. 2. Capture HTTP status, stable error code and request/correlation ID; never copy OTPs, passwords, access tokens or provider secrets. 3. Verify caller role/ownership and current authoritative state before retrying. 4. Check owning service health, then its provider/dependency after local/configuration causes are excluded. 5. If a release is implicated, compare artifact/config marker with prior known-good revision and validate rollback compatibility.

Data, security and deployment impact

The chef boundary owns its authoritative state; clients request or display changes but do not become the source of truth. Secrets and unnecessary personal data stay outside logs and PDFs. Releases must keep API, policy, schema and artifact compatible, with health/readiness evidence and rollback/backout appropriate to the change.

Evidence status

Implemented on main. Evidence: services/user-chef-service/README.md; apps/customer-web-next/src/components/chef/; apps/customer-web-next/src/components/chef-mode/. Snapshot: main @ 3225f7fa531a (14 August 2026).

chef state mismatch - errors and edge cases

Plain-English explanation

Chef mode is the supply side of the marketplace: onboarding, verification, menus, kitchen/order operations and earnings.

How it works technically

user-chef-service persists chef profiles and verification state. The frontend can switch context, but the server remains the authority: showing Chef mode never grants a CHEF role by itself.

Flow described in words

1. Actor starts the capability with the appropriate identity/context. 2. Client/BFF validates local prerequisites and sends an API request. 3. APIM and the owning service enforce identity, role, ownership and business-state rules. 4. The domain service reads/writes authoritative state and calls providers only through defined boundaries. 5. The caller receives authoritative result or a traceable error; support uses request/correlation evidence rather than secrets.

Errors, edge cases and controls

- Stale UI/cache must not override server state.
- UI visibility is not authorization evidence.
- Retries of money/state-changing operations require idempotency awareness.
- Provider/configuration outages should preserve core domain consistency where safe.
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catalog unavailable - plain-language purpose

Plain-English explanation

Catalog is the source of browseable meal and cuisine information used by discovery and commerce validation.

How it works technically

catalog-service is a Spring Boot/Flyway service with JWT controls. The web contains discovery, meals, search and selector modules. Administrative catalog operations are described by APIM guidance, including bulk pause/import and SHA-256 integrity checks.

Flow described in words

1. Actor starts the capability with the appropriate identity/context. 2. Client/BFF validates local prerequisites and sends an API request. 3. APIM and the owning service enforce identity, role, ownership and business-state rules. 4. The domain service reads/writes authoritative state and calls providers only through defined boundaries. 5. The caller receives authoritative result or a traceable error; support uses request/correlation evidence rather than secrets.

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Evidence status

Implemented on main. Evidence: services/catalog-service/README.md; apps/customer-web-next/src/components/discovery/; apps/customer-web-next/src/components/meals/; infra/apim/README.md. Snapshot: main @ 3225f7fa531a (14 August 2026).

catalog unavailable - technical architecture

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stale quote - plain-language purpose

Plain-English explanation

Checkout turns a basket into a validated purchase attempt and combines address, authoritative quote, order creation and hosted payment.

How it works technically

The browser does not decide the payable total. BFF/API calls obtain current backend state; Cashfree payment_session_id is used for hosted payment entry, while backend order/payment paths also document Razorpay acceptance and signed callbacks.

Flow described in words

1. Actor starts the capability with the appropriate identity/context. 2. Client/BFF validates local prerequisites and sends an API request. 3. APIM and the owning service enforce identity, role, ownership and business-state rules. 4. The domain service reads/writes authoritative state and calls providers only through defined boundaries. 5. The caller receives authoritative result or a traceable error; support uses request/correlation evidence rather than secrets.

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Data, security and deployment impact

The checkout boundary owns its authoritative state; clients request or display changes but do not become the source of truth. Secrets and unnecessary personal data stay outside logs and PDFs. Releases must keep API, policy, schema and artifact compatible, with health/readiness evidence and rollback/backout appropriate to the change.

Evidence status

Implemented on main. Evidence: apps/customer-web-next/README.md; apps/customer-web-next/src/components/cart/; apps/customer-web-next/src/components/checkout/; services/order-service/README.md. Snapshot: main @ 3225f7fa531a (14 August 2026).

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invalid order status - plain-language purpose

Plain-English explanation

Order management is the transactional spine of Craves: create, retrieve, status, cancellation/refund, delivery state, ETA/shortage, proof and administrative controls.

How it works technically

order-service documents idempotency, structured Problem Details errors, reconciliation, provider integration and controlled multi-database routing. Supported roles include customer, chef, admin and delivery-partner contexts with ownership enforcement.

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invalid order status - end-to-end flow

Plain-English explanation

Order management is the transactional spine of Craves: create, retrieve, status, cancellation/refund, delivery state, ETA/shortage, proof and administrative controls.

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order-service documents idempotency, structured Problem Details errors, reconciliation, provider integration and controlled multi-database routing. Supported roles include customer, chef, admin and delivery-partner contexts with ownership enforcement.

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ORDER_IDEMPOTENCY_CONFLICT - plain-language purpose

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